RESEARCH ARTICLE

Advanced Hybrid Machine Learning Model for Accurate Detection of Cardiovascular Disease

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Abstract
Cardiovascular disease (CVD) is one of the foremost reasons behind the death of people worldwide. Prevention and early diagnosis are the only ways to control its progression and onset. Thus, there is an urgent need for a detection model comprising intelligent technologies, including Machine Learning (ML) and deep learning, to predict the future state of an individual suffering from cardiovascular disease by effectively analyzing patient data. This study aims to propose a hybrid model that provides a deep insight into the data under consideration to enhance model accuracy for effectively detecting cardiovascular disease. This current research proposes a hybrid model comprising four stages. In the first stage of the proposed hybrid model, the data imbalance problem is solved using a hybrid sampling technique named Synthetic Minority Oversampling Technique-Edited Nearest Neighbors Rule. In the second stage, the Chi-square is applied as a feature selection method to select the highly relevant features from the records of 1190 with 11 clinical features, curated by combining the 5 most popular datasets, including Long Beach VA, Hungarian, Switzerland, and Statlog (Heart). In the third stage, the preprocessed dataset is passed to a stacking ensemble model comprising three base learners: Random Forest Tree (RFT), K-Nearest Neighbor (K-NN), and AdaBoost classifier and one meta-learner: Logistic Regression (LR), optimized with Grid Search Cross-Validation (GSCV) optimization approach, whose performance is evaluated against individual classifier. In the fourth stage, the performance is evaluated in terms of accuracy, sensitivity, specificity, F1 score, and ROC_AUC score.. The comparative results prove that the proposed hybrid model scored the highest accuracy of 97.8%, 96.15% sensitivity, and 96.75% specificity and 98.6% ROC_AUC score when compared with the existing techniques and models after applying the SMOTE–ENN (for data balancing) and Chi-square (for feature selection) methods for the efficient detection of cardiovascular disease. The implementation results demonstrate that the suggested hybrid model may accurately identify cardiovascular disease among patients. It facilitates the application of robust clinical treatment strategies.

Keywords Cardiovascular disease (CVD) · Machine learning (ML) · Synthetic minority oversampling technique-edited nearest neighbors (SMOTE–ENN)

1 Introduction

CVD is the primary reason behind the death of people worldwide. The global health report presented by WHO reveals that around "17.9 million people" lose their lives every year due to cardiovascular disease [1]. It mainly comes under the category of diseases that affect the blood and heart vessels. Medically, "atherosclerosis"

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causes the possibility of cardiovascular; it is a situation that arises due to the increase of plaque within the arteries [2]. Generally, plaque encompasses fatty deposits, cholesterol, calcium, and other substances that continuously harden and narrow the arterial walls and resist blood circulation to the brain, heart, and other vital organs, resulting in serious health issues [3]. Symptoms of cardiovascular disease vary from person to person, and the most common symptoms observed in cardiovascular patients include angina (discomfort), chest pain, breath shortness, fatigue, dizziness, palpitations, and slowness in ankles or legs [4]. However, in some cases, symptoms may not appear until any severe condition occurs, like stroke or heart attack. Along with these symptoms, some significant factors like high blood pressure, high cholesterol levels, diabetes, obesity, smoking, inactive lifestyle, and an ancestor's history of heart problems also cause cardiovascular disease [5, 6]. Therefore, it becomes essential to detect and handle the significant risk factors to identify and avoid the threat of cardiovascular disease [7].

Depending on these risk factors, doctors can suggest several preventative procedures that people at high risk of cardiovascular disease must follow [8]. Major preventive procedures include a healthy diet, regular exercise, involvement in physical activity, reducing stress, maintaining stable body weight, quitting smoking, avoiding salt, consuming fiber-rich food, taking proper medication, and limiting alcohol that will strengthen the arteries and clean the atherosclerotic plaque to reduce the possibility of cardiovascular disease [9].

Complex interactions among major risk factors generate various health issues, and conventional statistical methods cannot resolve all the complicated interactions between different risk factors [10]. Thus, monitoring the major risk factors by undergoing regular checkups to avoid any risk of developing cardiovascular disease becomes essential. However, manual checkups are very inconvenient and time-consuming. Therefore, an intelligent system must track patients' health conditions conveniently and effectively. Today, the Internet of Things (IoT) has shown great importance in the healthcare system and offers remote monitoring services to patients [11]. Smart wearable devices, such as bracelets and watches, monitor and collect accurate data regarding patient health [12, 13]. The growing usage of smart devices generates vast amount of data that must be effectively analyzed to gain better knowledge about patient health status.

Today, the healthcare field has a wide range of modern diagnostic tests that, when combined with Machine Learning (ML) [14], artificial intelligence, and different optimization approaches [15], become powerful tools for effective diagnosis and prevention of any disease [16, 17]. In addition, ML techniques help healthcare researchers identify significant diseases in a precise and effective manner [18]. Machine Learning (ML) plays a vital role in the early detection and prediction of various diseases, including diabetes, hypertension [17], breast cancer [19], Parkinson's disease (PD) [20], liver disease [21], chronic kidney disease [22], cardiovascular disease [23–26], lung cancer, etc.

With the increasing number of deaths due to cardiovascular diseases [27], it has become essential to propose a model for the efficient and reliable detection of cardiovascular disease. However, a big problem with healthcare institutions (hospitals, clinics, and healthcare centers) is the existence of large amounts of unmanaged data. With the growing utilization of innovative technologies in capturing patients' health data, significant issues like missing values, non-correlated features, data imbalance, noise, etc., may be associated with the data, leading to inefficient results [19, 20, 28–30]. This fact motivated us to work effectively on data collected from different sources before reaching the classification stage to get highly effective results. Therefore, by considering all these significant issues, the current work focuses on proposing a cardiovascular disease detection model using effective ML techniques and making keen observations of the data under consideration.

Significant contributions of this work are as follows:

1. The most crucial step in the proposed hybrid model is data pre-processing, computed by data balancing and cleaning techniques. Data balancing is performed using a hybrid balancing approach called SMOTE–ENN. Here, SMOTE is an oversampling method used to enhance minority samples, and ENN is an undersampling method used to reduce noise generated after SMOTE. By following this method, a balanced dataset is obtained for training the model.

2. After that, a feature selection method (Chi-square) is applied to select the most relevant features, aiming to reduce computational time and improve model performance.
3. After feature selection, the selected set of features is passed to a stacking model with three base learners: RFT, K-NN, and AdaBoost classifier, and one meta-learner, LR. Finally, the proposed hybrid model and other existing techniques are compared using accuracy, specificity, sensitivity, precision, F1_score, and ROC_AUC as evaluation metrics.

The primary outlines of this paper are as follows: Sect. 2 provides an overview of the work done by different researchers related to CVD detection and prediction. Section 3 details the model proposed in this paper. Section 4 discusses the applied dataset, experimental results, and analysis. Lastly, the conclusion of this work is given in Sect. 5.

2 Related Work

Researchers have suggested various practical methods for identifying cardiac diseases using different M techniques.

Subramani et al. designed a model that used the Gradient Boosting Decision Tree (GBDT) approach with Shapley additive explanations (SHAP) for feature selection. They integrated different classification techniques (KNN, DT, NB, SVM, RF, LR, and XGB) for the prediction of CVD and found that a stacking-based classifier achieved the best accuracy [31].

Ahmad et al. stated that ML techniques play a significant role in the healthcare domain for disease diagnosis, especially for cardiovascular detection. The author performed a comparative analysis of 12 commonly used ML techniques: RF, GB, DT, KNN, NN, LR, Quantitative Descriptive Analysis (QDA), Nu Support vector classifier (NuSVC), Linear Discriminant Analysis (LDA), Naïve Bayes, SVM, and a stacking classifier. A ROC_AUC score of 100% was achieved for RFT and GB techniques [32].

Abdullah et al. developed an ensemble learning model by integrating machine and deep learning techniques to detect cardiovascular disease. The final prediction of the class label was decided by the majority voting of DNN, KDNN, and stacking of XGB, KNN, and DT, with an accuracy of 88.70. However, the accuracy is very low and can be further improved [33].

According to Dritsas Trigka, ML is gaining popularity in predicting various diseases. The author followed the SMOTE to balance the given dataset. The performance of different ML techniques was evaluated and analyzed in terms of recall, accuracy, area under the curve, and precision. Findings state that out of all methods, the stacking ensemble (classifier: RFT, NB, and meta classifier: Logistic Regression (LR)) attained the highest accuracy score of 87.8%, which can be further improved using feature selection methods.[34].

Trigka and Dritsas proposed an ML model for predicting major risk factors for coronary artery disease. The authors applied various ML models after balancing the data using the SMOTE method for experimentation. The importance of features is also considered before training the model. The suggested model attained 90.9% accuracy and 96.1% ROC_AUC [35].

Amin et al. worked on identifying significant features in heart disease prediction. The author applied seven different ML techniques, i.e., NB, DT, LR, K-Nearest Neighbor (K-NN), NN, SVM, and the voting method (hybrid LR and NB), to different combinations of features and found the voting classifier to be the best model [36].

Martins et al. developed an optimized Decision Tree model for the identification of heart disease. The model's outcome was also compared with other ML techniques (DT, RF, DL) and showed the highest accuracy of 73.58% [37].

Kamel et al. developed a hybrid RDO (Random Drift Optimization)–XGBoost approach to optimize the identification of breast cancer. Along with improving classification performance, the proposed approach also provides valuable insights into the most significant features deriving the progression of breast cancer. However, the

proposed approach acts as a powerful strategy for advancing feature selection in cancer research. Still, additional research is required to validate their findings on the more extensive and diverse dataset [38].

Yaqoob et al. introduced a deep learning model for the detection of breast cancer by incorporating the hybrid selection approach by combining "Harris Hawks Optimization (HHO) and Whale Optimization (WO)" algorithms [39]. The computed results reveal that the proposed model outperformed other existing approaches and attained a mean classification accuracy of 99.0%, which facilitates individual treatment and thus improves the patient's quality of life. Although the model shows better results, its performance can also be tested for the diagnosis of another disease and on some sensors-based datasets.

Yaqoob et al. found out that the diagnosis of breast cancer largely depends on the gene expression of patients. However, the existence of irrelevant and redundant genes can make the diagnosis more challenging. To overcome this problem, the authors proposed a novel feature selection approach by combining the HHO and cuckoo search algorithm (HHOCSA). The efficiency of the proposed approach was examined by implementing it on six different datasets. The experimental results demonstrated that the developed HHOCSA approach is more efficient and consistent in terms of variability and outperforms other existing approaches in the identification of breast cancer. However, the proposed approach is highly significant in identifying breast cancer; therefore, the same approach can be applied to the diagnosis of other diseases and identifying the considerable patterns from complex datasets [40]

Budholiya et al. used a Bayesian optimization approach for the XGBOOST (Extra Gradient Boosting Decision Tree) classifier to predict heart disease. Following a one-hot encoding scheme, the proposed model was tuned with hyperparameters. The outcome of the developed model was compared with the RF and ET (Ensemble Tree) classifiers. The developed model attained an accuracy score of 91.8%, which can be further improved with ensemble classifiers. [41].

Feshki and Shijani used the Particle Swarm Optimization (PSO) approach to improve the performance of the applied model (neural network). PSO was followed after pre-processing to rank the features. The original dataset contains 13 features, out of which PSO selected 8. Findings show that the proposed "hybrid PSO-FFBP (hybrid-particle swarm optimization-feed forward back propagation)" model attained an accuracy of 91.94% for heart disease prediction [42].

Haq et al. introduced a hybrid model for predicting heart disease by incorporating three different approaches: i.e., Least absolute shrinkage and selection operator (LASSO), Relief FS, and Maximum relevance minimum redundancy (mRMR) before passing to any ML model (LR, K-NN, ANN, NB, SVM, DT, and RF). The evaluated results showed that the model attained a better accuracy score with the Relief FS method. The highest accuracy achieved was 88.7%, which can be further improved. Along with that, the size of the dataset is tiny [43].

Previous research has revealed that cardiovascular disease is one of the most life-threatening diseases, and additional research on cardiovascular disease must be conducted to find an effective solution that could help develop a smart healthcare system. However, all the above-stated methods are limited only to applying ML algorithms without balancing or observing data; unbalanced data may provide biased results. So, the current work focuses on developing an effective cardiovascular disease detection model that keenly observes data before making any prediction.

3 Proposed Work

Figure 1 thoroughly represents the proposed hybrid model, comprising four stages. In the first stage of the proposed hybrid model, the data imbalance problem is solved using a hybrid sampling technique named SMOTE–ENN. In the second stage, the Chi-square is applied as a feature selection method to select the highly relevant features from the records of 1190 with 11 clinical features, generated by combining the 5 most popular datasets, including Long Beach VA, Hungarian, Switzerland, and Statlog (Heart). In the third stage, the preprocessed dataset is passed to a stacking ensemble model comprising three base learners: Random forest tree (RFT), K-NN, and

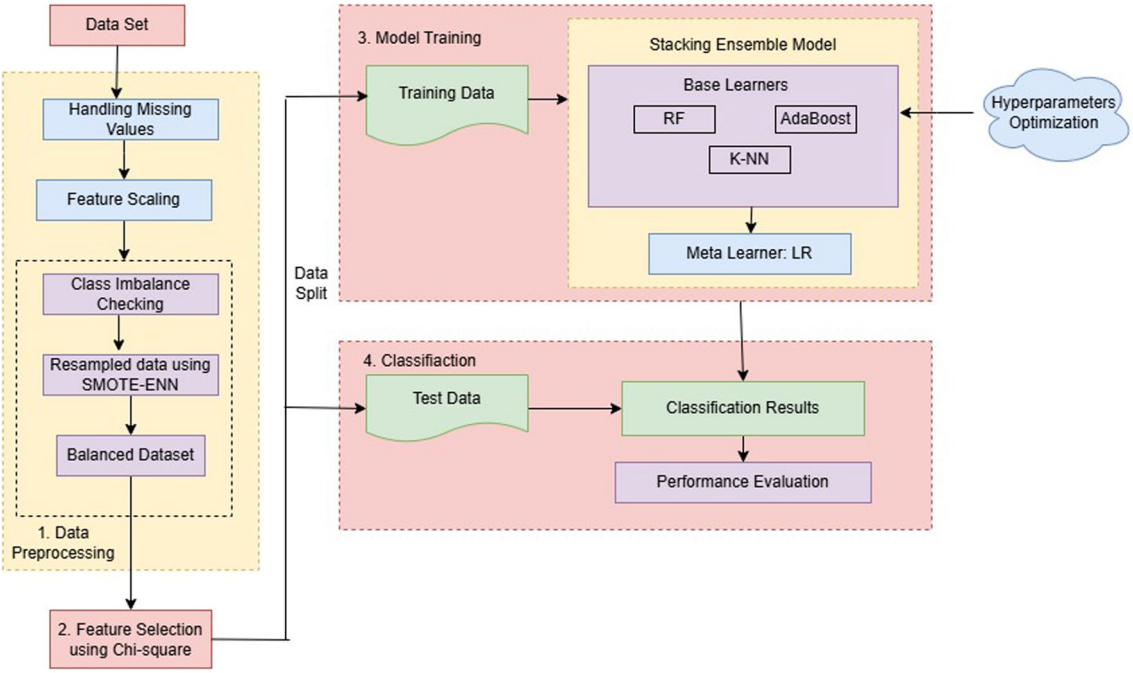


Fig. 1 Diagrammatic view of the proposed hybrid model for cardiovascular disease detection

AdaBoost classifier and one meta-learner: LR whose performance is evaluated against individual classifier [21, 44–47]. In the fourth stage, the performance is assessed in terms of accuracy, sensitivity, specificity, F1 score, and ROC_AUC score.

The ML techniques for the proposed hybrid model are selected by analyzing their performances after being implemented on the same dataset, as described in Table 5.

3.1 Dataset

In the current work, the publicly existing dataset from the IEEE data port repository "(https://ieee-dataport.org/open-access/heart-disease-dataset-comprehensive)" is utilized, which consists of a total of 1190 records with 11 clinical features and 1 target column, curated by combining the 5 most popular datasets including Long Beach VA, Hungarian, Switzerland, and Statlog (Heart). The target column refers to the occurrence of cardiovascular disease or not among individuals [23–27, 48]. Table 1 gives a detailed explanation of the collected dataset's features, consisting of 12 features, out of which 11 are the clinical features and 1 is the target column. Clinical features include the patient's age, sex, type of chest pain, resting bps, fasting blood sugar, maximum heart rate, cholesterol, resting ECG, ST slope, old peak, exercise again, and the target column describes the status of the patient whether they have cardiac disease or not. In addition, it also represents the feature type, whether numerical or categorical.

After the dataset is selected, it is thoroughly checked and then preprocessed accordingly to achieve the best classification results, as described in the next section.

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Table 1 Feature description of dataset [49]

No	Feature_Name	Description of features	Feature_Type
1	Age	Age of persons in Yrs	Number
2	Sex	0: Female, 1: Male	Category
3	Chest pain type	Type of pains(1: typical angina, 2: atypical angina, 3: non-anginal, and 4: asymptomatic)	Category
4	Resting bps	Blood pressure in (mm-Hg)	Number
5	Fasting blood sugar	No: 0, Yes: 1	Number
6	Cholesterol	Cholesterol in (mg/dL)	Category
7	ECG resting	Results of ECG (0: normal, 1: ST-T wave abnormality, 2: left ventricular hypertrophy)	Category
8	Max heart rate	Maximum heart rate of patient	Number
9	Exercise angina	Yes: 1, no: 0 (angina due to exercise)	Category
10	Oldpeak	Level of peak exercise	Number
11	ST slope	Level of peak exercise ST segment (1: upsloping, 2: flat, 3: down sloping)	Category
12	Target	Status of cardiac disease	Category

3.2 Data Preprocessing

To get effective results, it is first required to check whether the dataset contains missing, unnormalized or unbalanced values. Therefore, to handle any missing and unnormalized dataset, it is first passed to the data cleaning phase.

3.2.1 Data Cleaning

In the data cleaning phase, the dataset is first checked for missing values and whether it contains any missing values or not. To check this, the ".isna()" method is computed on the collected dataset in Python. From the computed results, it is found that no missing value exists. After that, to check the distribution of data along each feature, a visual representation of data is computed using the "histplot()" method in Python [50–53]. The histogram plot obtained after the computation of the "histplot()" method is shown in Fig. 2.

Histplot displays the distribution of each feature in the dataset. Each histogram represents a different feature, such as age, cholesterol levels, sex, systolic blood pressure, resting blood pressure and other health-related

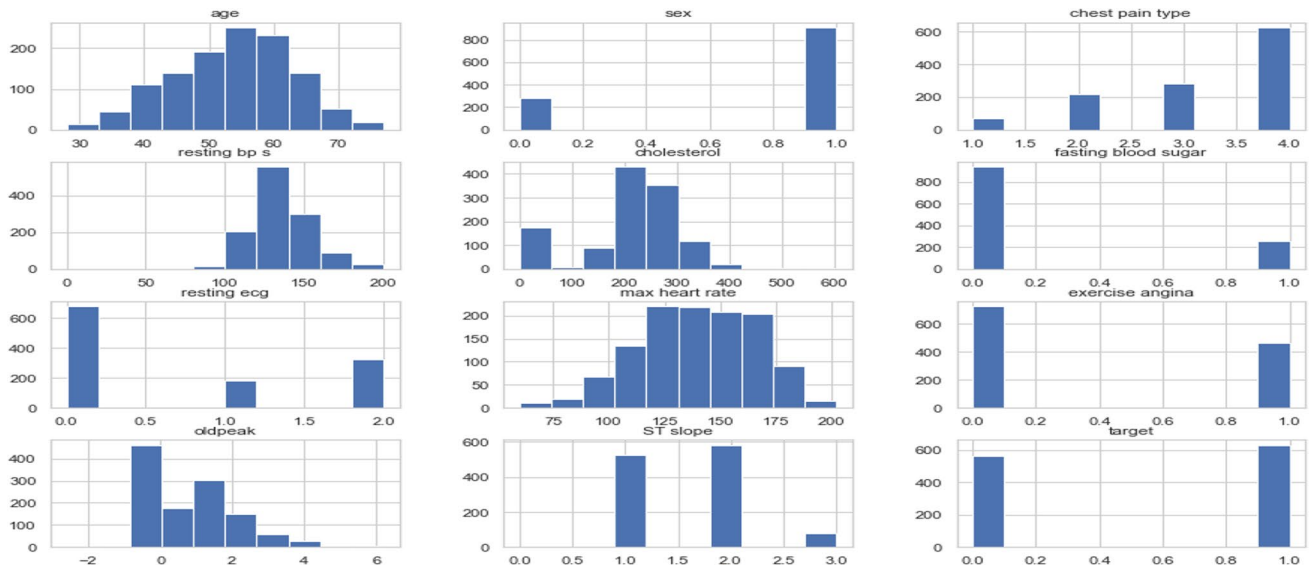


Fig. 2 Histplot illustrating the distribution of features in the selected dataset

metrics. The x-axis of each histogram shows the feature values (e.g., age in years, cholesterol in mg/dL), while the y-axis indicates the frequency of occurrences. Figure 2 shows uneven disturbances in some features, including heart rate, cholesterol, age, and blood pressure. Such uneven disturbances can cause inaccurate results while training the model. Therefore, the min–max scaling method is followed to normalize the data during the pre-processing phase to obtain a normal distribution. Features are transformed by scaling each feature to a given range and computed using Eq. 1.

$$\frac{y_i - \min(y)}{\max(y) - \min(y)} \tag{1}$$

It transforms features by scaling each one to a specified range, usually $[-1, -1]$ or $[0, 1]$. Figure 3 represents the distribution of features after applying the min–max scaling approach.

Figure 3 depicts the features normalized after the min–max scaling approach so that each feature contributes equally to the analysis, preventing features with larger magnitudes from dominating those with smaller ones.

3.2.2 Data Balancing

After feature scaling, we check whether the collected data are balanced or not. Therefore, the distribution of data corresponding to its target column is checked. It is observed that there is a significant variation between class samples. If we train the model with this imbalanced data, our model may be biased towards the majority class. In addition, the imbalanced dataset can bias the performance metrics, such as accuracy. Therefore, an efficient method must be followed to balance the dataset to avoid biases and to improve model performance. In the proposed model, we have observed a hybrid sampling (SMOTE–ENN) approach to balance the collected dataset [50, 54, 55]. The selection of the SMOTE–ENN approach is based on the results achieved after comparing the performance of different data balancing approaches, presented in Table 2. Though different data balancing approaches exist among all, SMOTE, SMOTE–ENN, and Random Oversampling are the most popularly used for handling the data imbalance problem. Therefore, to select the best data balancing approach, we have implemented these data

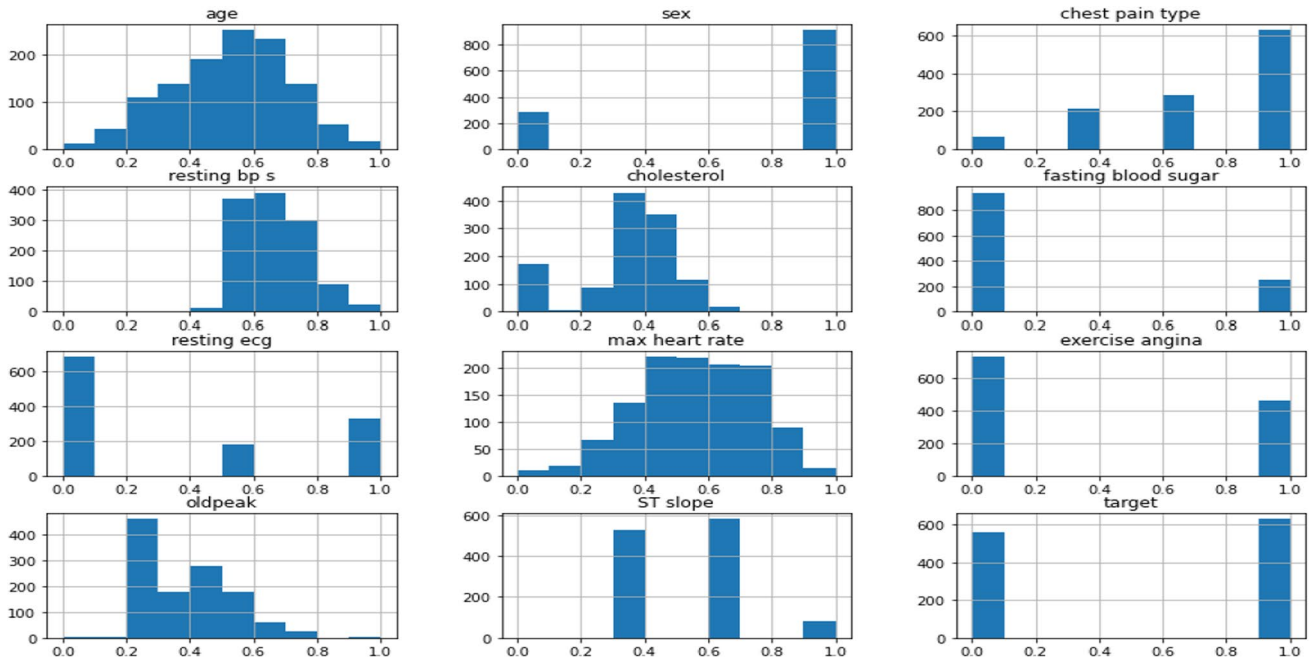


Fig. 3 Histplot illustrating the distribution of features in the selected dataset after the min–max scaling approach

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Table 2 Performance results in terms of accuracy of different data balancing approaches

	Random Oversampling	SMOTE	SMOTE–ENN
LR	83.73	85.71	**88.24**
DT	90.48	88.10	**92.51**
RFT	92.06	91.25	**93.25**
AdaBoost	88.10	86.51	**88.24**
K-NN	90.48	89.68	**93.05**
NB	86.11	86.11	**88.77**
Stacking Ensemble	92.66	92.46	**97.8**

balancing approaches on the chosen dataset using different ML techniques discussed above. Their performance is analyzed in terms of accuracy, as listed in Table 2.

From the results shown in Table 2, it can be concluded that all data balancing approaches provide effective results. Still, the SMOTE–ENN approach shows highly efficient results in terms of accuracy compared to other methods. Therefore, in the proposed model, the SMOTE–ENN approach is used to handle the data imbalance problem. The algorithm followed by the SMOTE–ENN approach is described below:

Algorithm 1 Hybrid sampling algorithm (SMOTE-ENN)

Input: Training set T_{train}, S no. of nearest neighbor in SMOTE, N no. of nearest neighbors in ENN
Output: New training set Ttrain_new: Obtained after SMOTE-ENN
Divide Ttrain into positive and negative samples
Ttrain= Pos Ù Neg ;
Oversample the minority class using SMOTE
New_Pos<-SMOTE(Pos,S) and
|New_Pos|=|Neg|;
T_{train_new}=New_Pos Ù Neg;
For xεT_{train_new}do
|Compute distance according to S|
End
Remove noisy samples using ENN based on N –nearest neighbor distance
T_{train_new}<-ENN(T_{train_new}, S, N)

The SMOTE–ENN approach combines the features of both undersampling and oversampling techniques to improve the quality of the training data. In this approach, initially, the SMOTE method increases the samples of minority classes to balance the data. However, the resulting dataset may contain noise and misclassified samples. Therefore, after generating synthetic samples, the ENN approach is applied to remove ambiguous and noisy samples, improving the overall quality of the training dataset. Results achieved after implementing the SMOTE–ENN method are shown in Fig. 5(b).

3.3 Feature Selection (FS)

In our proposed model, we employed Chi-square [51]. As an FS method to identify the most significant features to improve model performance, this method works on the principle of finding the independent set of features against the target class and assigning a value, which will demonstrate their degree of correlation. This test is applied for every positive value to find the features on which the targeted class depend [52]. In the current study, the problem of binary classification is considered, and for any binary classification problem, the Chi-square is computed by assuming a total (t) number of instances for both negative and positive classes. Table 3 represents the values used for computing the Chi-square for each feature.

Here in Table 3, m is the summation of positive instances; (m-t) denotes the summation of negative cases; in the same way, w specifies the sum of instances when (yi) exists, and (t-w) shows the sum of cases where (yi) is not present. “Chi-square” compares the actual count or original (O) with the predicted or expected count (E). When the

Table 3 Assumptions used for computing Chi-square value

	+ve	−ve	Total
When feature y_i is present	P	Q	w = p + q
When feature y_i is absent	R	S	t-w = r + s
	m = p + r	**m-t = q + s**	**T**

features are independent, the values of both counts are very close to each other. E_p, E_q, E_r, and E_s denote the predicted values, and p, q, r, and s represent the values to be measured. E_p is computed using Eq. 2. Similarly, E_q, E_r, and E_s are calculated. Equation 4 is used to compute the Chi-square score, while Eq. 3 is used to calculate the actual test.

$$E_p = (p+q) \times \frac{(p \times q)}{t} \tag{2}$$

$$X^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i} \tag{3}$$

$$X^2 = \frac{(p-E_p)^2}{E_p} + \frac{(q-E_q)^2}{E_q} + \frac{(r-E_r)^2}{E_r} + \frac{(s-E_s)^2}{E_s} \tag{4}$$

Figure 5 in the Result and Discussion section represents the importance score assigned to different features after applying the Chi-square approach.

3.4 Machine Learning Techniques

After the pre-processing phase, the most appropriate ML technique is selected among different popularly used ML techniques for efficient data analysis. Therefore, in this section, all the popularly used ML techniques are described, and then their performance is analyzed to select the most appropriate among them.

3.4.1 Logistic Regression (LR)

LR is a popular ML method that is appropriate for both regression and classification problems. This method is highly applicable to binary classification problems [56]. For binary classification problems, a "logistic function" is computed to know the class of new input using Eq. 5:

$$\textbf{Logit}\,(\mathbf{p}) = \frac{\textbf{prob. of the presence of characteristics}}{\textbf{prob. of the absence of characteristics}} = \textbf{In}\left(\frac{\mathbf{p}}{1-\mathbf{p}}\right) \tag{5}$$

3.4.2 Random Forest Tree (RFT)

RFT is an ensemble ML approach that builds multiple DTs based on randomly selected training data and then aggregates the results to make a final prediction. Such a technique is highly effective for both classification and regression tasks. RF resolves the decision tree's affinity for overfitting toward training data [57]. A majority voting approach is followed for classification tasks. Let C^i(y) be the class of ith DT. Then the prediction for any new data point is calculated using Eq. 6.

$$\widehat{\mathbf{Ci}}(\mathbf{y}) = \textbf{majority voting}\{\widehat{\mathbf{Ci}}(\mathbf{y})\}_1^{\mathbf{M}} \tag{6}$$

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3.4.3 Decision Tree (DT)

DT follows a tree-like structure to make predictions and decisions. It is the most popular technique used for healthcare data analysis. A DT is built from a given training dataset that is used to extract the set of rules, which helps in evaluating the class and label of a given instance. It is composed of branches, nodes, and leaves, where nodes represent features and attributes, branches specify the decision rules, and leaves specify the target variable or the outcome.

For splitting at each node, Info Gain and Entropy are used as splitting criteria, and they are calculated using Eqs. 7 and 8.

$$\textbf{Entropy } (\mathbf{H(C)}) = -\sum_{\mathbf{i=1}}^{\mathbf{n}} \mathbf{p_i log_2 p_i} \tag{7}$$

$$\textbf{InfoGain } (\mathbf{C}, \mathbf{F}) = \mathbf{H(C)} - \mathbf{H(C|F)} \tag{8}$$

Here, H(C) is the entropy, and H(C|F) refers to the conditional entropy of class variable "C" for specified attribute F. Here, the number of classes is denoted by n, and pi denotes the relative probability of iεC.

3.4.4 K-Nearest Neighbor (K-NN)

It is a powerful ML technique highly applicable to labeled datasets. In such an approach, a new data point is classified based on the distance of data that K-NN classifies. In the K-NN approach, K denotes the number of nearest neighbors to be used to predict the new data points. The value of K is selected based on the smallest distance from the input test sample assigned to the class with the highest occurring frequency of K-NN. Proximity is calculated using Euclidean distance (ED) using Eq. 9.

$$\textbf{Euclidean distance} = \sqrt{\sum_{\mathbf{i=1}}^{\mathbf{N}} (\mathbf{ai} - \mathbf{bi})^2} \tag{9}$$

3.4.5 Naïve Bayes (NB)

NB is an ML technique that works on the principle of Bayes theorem. Such a technique is highly effective for problems with data from multiple dimensions. Using this approach, the probability of any feature F belonging to class C_i is calculated using Eq. 10 as follows;

$$\mathbf{P}\left(\mathbf{C_i}|\mathbf{F}\right) = \frac{\mathbf{P(C_i)P}\left(\mathbf{F}|\mathbf{C_i}\right)}{\mathbf{P_F}} \tag{10}$$

Here, $P(C_i)$is prior probability, C_i|F: is the posterior probability, P(F) is evidence, and $P(F|C_i)$ is a likelihood.

3.4.6 AdaBoost

AdaBoost is an adaptive boosting method that comes under an ensemble learning approach that is primarily useful for binary classification but can be extended to multi-classification. It works on the principle of combining multiple weak techniques to generate a robust method. Initially, equal weight is assigned to each training instance. After that, the weak technique is trained on these weighted training data to evaluate the

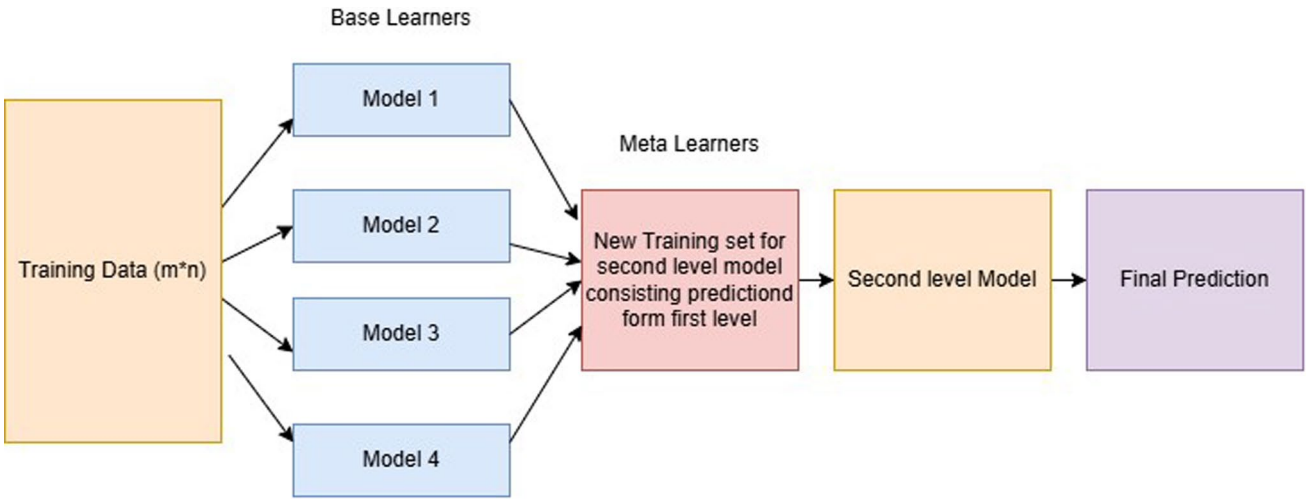


Fig. 4 Workflow depicting the steps followed by stacking ensemble technique [58]

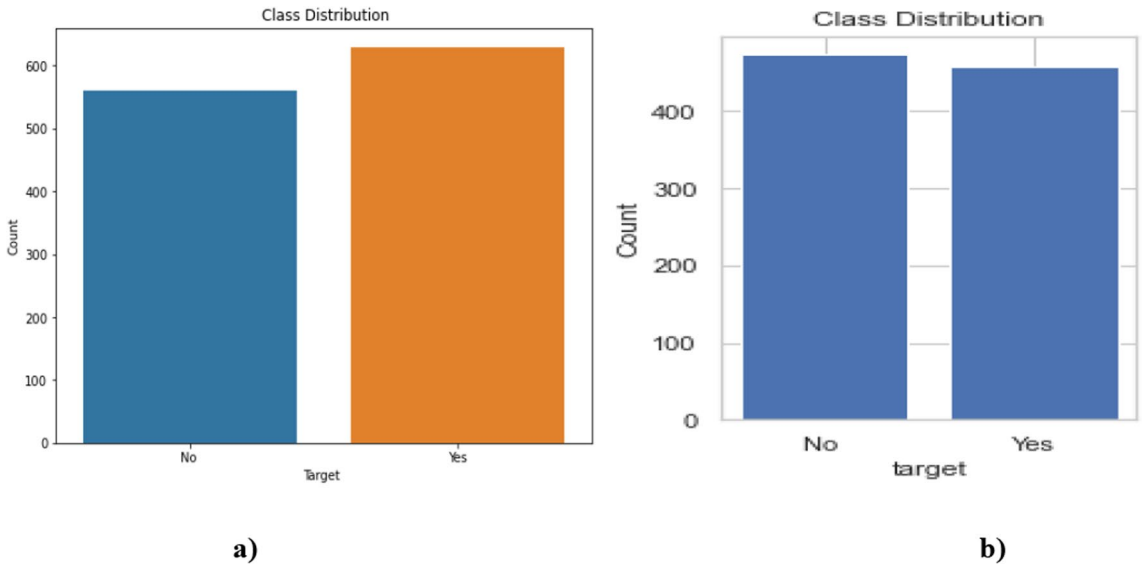


Fig. 5 Depicts the distribution of target datasets in different classes before and after the SMOTE–ENN approach. **a** Before SMOTE–ENN **b**After SMOTE–ENN

classification error. At each phase, the weights of accurately classified instances are decreased while the mistakes of wrongly classified instances increase. The final result is the weighted sum of all weak classifiers.

3.4.7 Stacking Ensemble

Stacking techniques come under the ensemble learning approach. This technique consists of two or more base/learners' techniques and one meta-learner that combines the prediction made by base learners. Base learners work at zero level, and meta-learners work at level one. In the proposed technique, the stacking ensemble technique is generated using three base learners: RFT, K-NN, and AdaBoost at level zero and one meta-learner, LR, at level 2 of the stacking technique [19, 20, 28–30]. The basics of the stacking ensemble technique are shown in Fig. 4.

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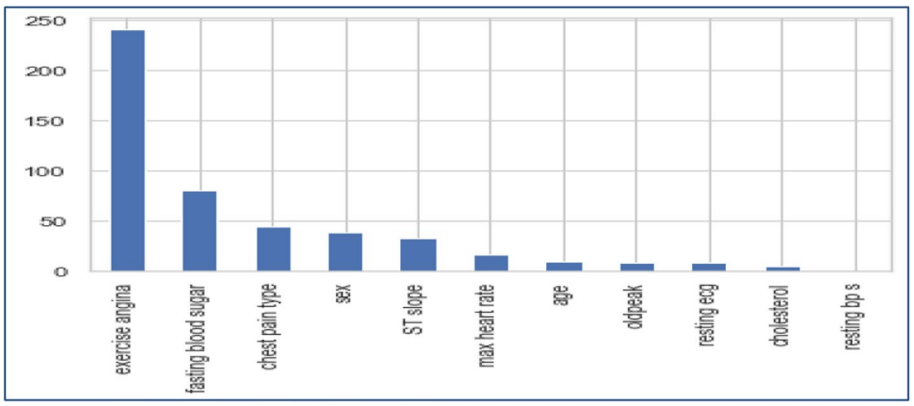


Fig. 6 Illustrates the importance score assigned to different features by the Chi-square approach

Table 4 Evaluation metrics used to evaluate performance of the proposed hybrid model

Metrics_Name	Formula	Description
Accuracy	$(T_P + T_N)/T_N$	It is the ratio of truly predicted CVD patients out of the total patients
Sensitivity	$T_P/(F_N + T_P)$	The ratio of truly CVD patients that are predicted as truly CVD
Specificity	$T_P/(F_P + T_P)$	The ratio of truly positive samples that were predicted as true
F1_score	2(recall * precision)/(recall + precision)	The harmonic means of both recall and precision

4 Results and Discussion

In the current study, the proposed hybrid model shown in Fig. 1 is implemented to diagnose cardiovascular diseases using the SMOTE–ENN class balancing approach, with Chi-square as an FS method, and a stacking model with three base learners: RFT, K-NN, and AdaBoost classifier and one meta-learner: LR, based on extensive comparison and analysis results, as described below. The SMOTE–ENN approach is applied to the CVD dataset, which consists of the five most popular datasets, including "Hungarian, Long Beach, Switzerland, VA, and Statlog (Heart)." Fig. 5 represents the bar plot for a better understanding of the distribution of classes among CVD and non-CVD patients. It represents the class distribution before and after applying SMOTE–ENN (a data balancing approach). Figure 5(a) shows an imbalanced data distribution between CVD and non-CVD classes. Therefore, a balancing approach must be needed to balance the dataset. In our work, we have applied the SMOTE–ENN approach to balance the dataset, whose computed results are represented in Fig. 5(b).

After data balancing, the Chi-square is applied to the dataset to rank all 11 features in descending order according to their importance score, as shown in Fig. 5.

Figure 6 shows the feature importance score assigned by the Chi-square method. From Fig. 6**,** it is observed that features like fasting blood sugar, exercise angina, sex, St slope, chest pain type, and max heart rate show high ranking scores, while features like age, old peak, resting ECG, cholesterol and resting bps. Therefore, these low-ranked features are removed before training the model, thus improving model efficiency and reducing complexity. A threshold of 5 features is set, which reduces the features from 11 to 5, thereby reducing computational time and improving model efficiency. Next, the collected dataset is partitioned into a ratio of 70:30, in which 70% of the data is used for training, and 30% of the data is used for testing the model.

The proposed model is evaluated based on four major parameters: Tp denotes that predictions for patients with CVD are accurate, Fp denotes that patients without CVD are wrongly classified as having CVD, Fn denotes that patients with CVD are improperly classified as healthy, and Tn denotes that prediction for people without CVD is accurate. Different performance evaluation measures, including accuracy, sensitivity, specificity, and F1 score, that were used to implement the proposed model are described in Table 4.

Table 5 Parameters passed to different ML techniques

Techniques	Hyper_Parameter	Value
Random forest	max_ depth	10
	n_estimators	100
	min_ samples _split	2
	Criteria	Entropy
DT	Criteria	Entropy
	Max_depth	16
Logistic regression	C	1.0
	Penalty	12
	Solver	Liblinear
Naïve Bayes	–	–
K-NN	n_neighbors	7
	Weights	Uniform

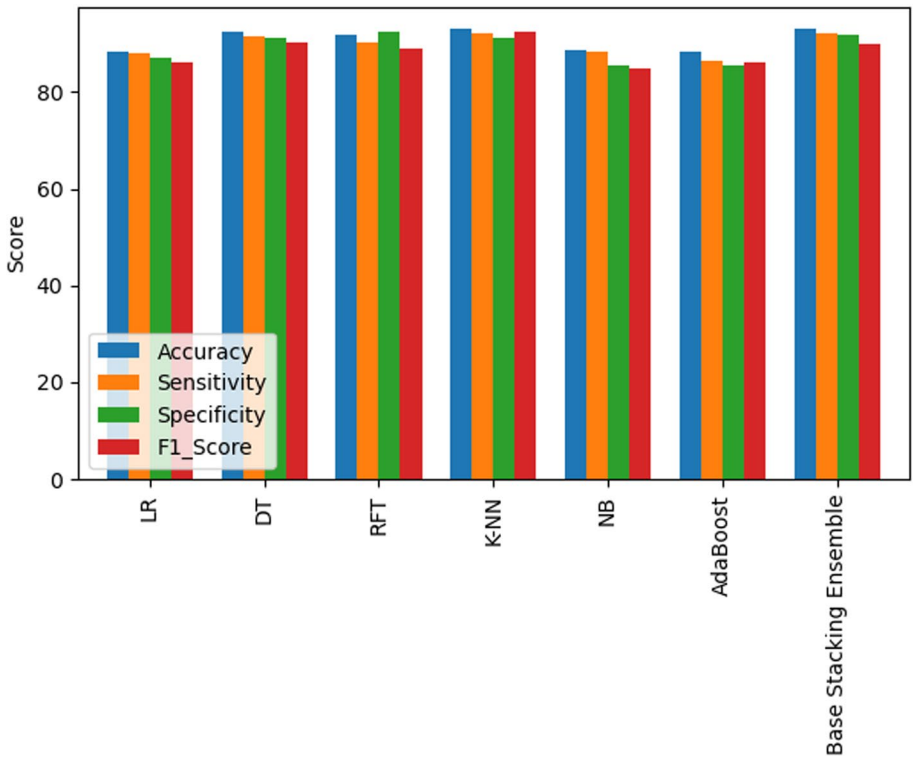


Fig. 7 Illustrating a comparison between the performance of different ML techniques and the base stacking ensemble model without class balancing and FS approach

The proposed hybrid model outperforms when implemented on the given dataset compared to the individual ML techniques, as shown in Table 7 and Fig. 8. Before implementing the proposed model, the individual ML techniques are implemented on the given dataset without applying FS and the data balancing approach, whose results are listed in Table 5 and Fig. 7. Each technique is tuned with hyperparameters optimized through the Grid Search Cross-Validation (GSCV) method using the tenfold CV method. GSCV is the most popular technique for optimizing hyperparameters; it involves extensive search through a specified parameter grid and thus evaluates the model performance for every possible combination of hyperparameters. The selected set of hyperparameters for individual techniques is listed in Table 5.

From the computational results listed in Table 6, it is observed that RFT and K-NN performed well among all base classifiers and achieved 93.70% and 92.86% accuracy scores. LR and NB achieved the lowest accuracy scores of 84.03% and 85.75%, respectively. Further, to improvise the results, a base stacking ensemble model

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Table 6 Performance comparison between different ML techniques and the base stacking ensemble model without class balancing and FS

ML models	No. of features	Accuracy	Sensitivity	Specificity	F1 score	ROC_AUC
LR	12	84.03	84.0	84.37	84.17	93.2
DT	12	86.97	85.17	86.5	85.47	87.3
RFT	12	93.70	93.5	93.6	92.76	97
AdaBoost	12	89.08	88.0	89.07	88.97	96.6
K-NN	12	92.86	92.85	92.38	92.45	93.0
NB	12	85.75	85.73	85.47	85.37	94.3
Base stacking ensemble model	12	94.74	94.45	93.25	93.78	97.2

Table 7 Comparison between the performance of different ML techniques and the proposed hybrid model after the application of class balancing and FS approach

ML models	No. of Features	Accuracy	Sensitivity	Specificity	F1 score	ROC_AUC
LR	5	88.24	88.14	87.25	86.14	96.3
DT	5	92.51	91.45	91.30	90.45	93.5
RFT	5	91.98	90.14	92.45	89.05	98.2
K-NN	5	93.05	92.35	91.15	92.67	97.9
NB	5	88.77	88.45	85.65	85	96.1
AdaBoost	5	88.24	86.45	85.4	86.15	94.8
Proposed hybrid model	5	97.8	96.15	96.75	95	98.6

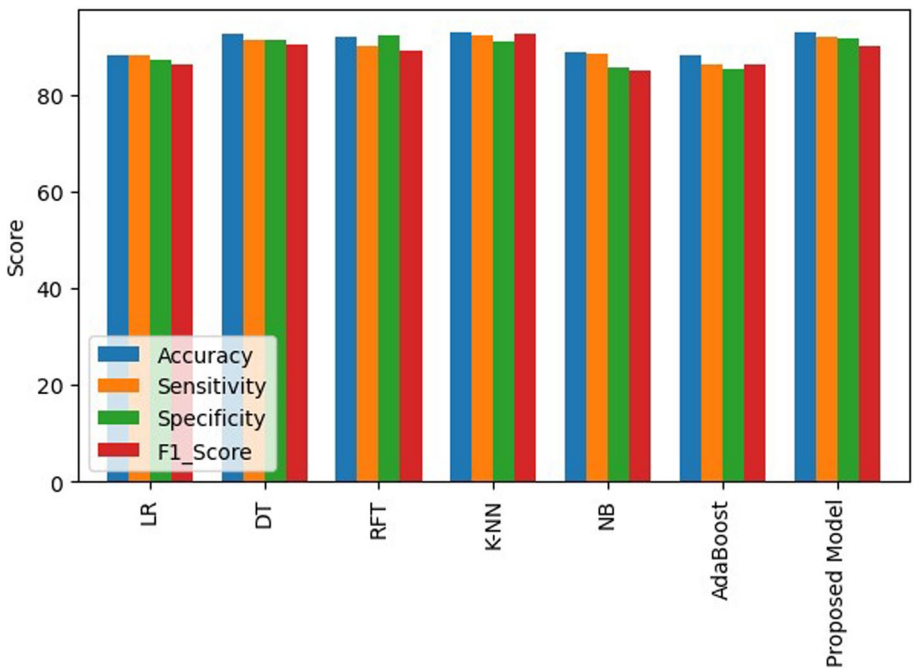


Fig. 8 Illustrating the comparison between the performance of different ML techniques and the proposed hybrid model with class balancing and FS approach

is developed with three base learners: RFT, K-NN, and AdaBoost classifier and one meta-learner, LR, on the basis of the results of the individual ML techniques, which outperformed the other ML techniques and scored an accuracy of 94.74%, showing a notable improvement of 1.04% in accuracy.

Further, to improve the results, the proposed model with efficient data balancing and FS approach is implemented on the given dataset and tuned with hyperparameters, as listed in Table 5, using the tenfold CV method.

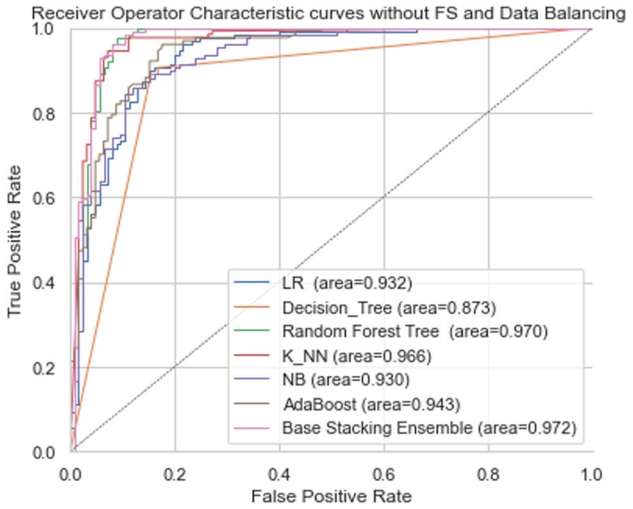


Fig. 9 ROC-AUC curve depicting the performance of different ML techniques and the base stacking ensemble model without SMOTE–ENN and FS approach

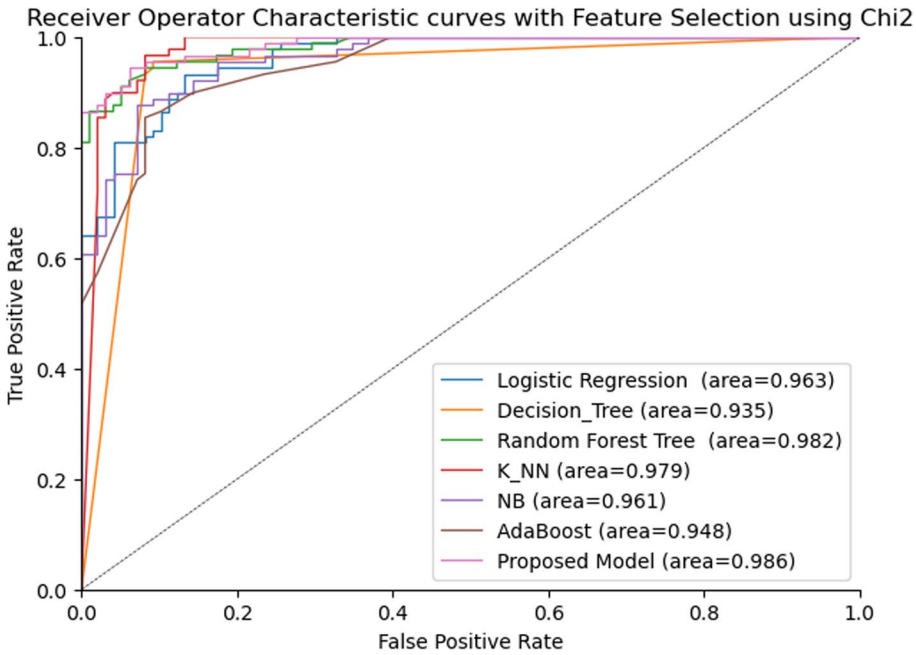


Fig. 10 ROC-AUC curve depicting the performance of different ML techniques and the proposed hybrid model with SMOTE–ENN and FS approach

The computational results achieved after the implementation of the proposed model are listed in Table 7 and graphically presented in Fig. 8. Table 7 shows that the proposed hybrid model attained the best accuracy score of 97.8% in comparison to the other ML approaches.

Further, the proposed hybrid model is evaluated using the ROC-AUC chart, in which the ROC-AUC score is shown on the x-axis, and false positive rates are shown on the y-axis. This curve will check a model's ability to significantly classify instances into two classes: "no CVD" and "CVD." We further evaluate the performance of different ML techniques and the proposed hybrid model using the ROC-AUC curve for detecting CVD, as shown in Figs. 9 and 10.

The computational results show that all ML models achieved high performance after applying the SMOTE–ENN and FS techniques. Our implementation's results state that the proposed hybrid model attained the highest accuracy score.

Table 8 illustrates a comparative analysis of the proposed hybrid model and existing models applied to the same dataset. The comparative analysis of the current work with the previous work is done based on the methodology

Table 8 Comparative study of proposed hybrid CVD prediction model with existing models

Author's	Methodology	Results
Mohan et al. [59]	Feature selection + hybrid approach (RF + Linear Model (LR))	Achieved accuracy is 88.70%
Martins et al. [37]	Optimized DT, DT, RI, RF, DL	Optimized DT achieved the highest accuracy of 73.58%
Alqahtani et al., 2022, [33]	LR, DT, RF, KNN, NB, SVM, MLP	No SMOTE and feature selection, ML ensemble achieved the highest accuracy of 88.70%
Trigka et al. [35]	SMOTE + feature selection + (3NN, Bagging, J48, LR, MLP, NB, RF, RotF, voting and stacking)	Stacking ensemble achieved the highest accuracy of 90.9%
Proposed hybrid model	SMOTE_ENN + feature selection + stacking(LR, RFT, K-NN, AdaBoost)	Achieved the highest accuracy of 97.8% and ROC_AUC score of 98.6%

used by different researchers in detecting cardiovascular disease. In most previous studies, only the SMOTE method was used for data balancing with traditional ML techniques. In the current study, in the proposed hybrid model, we have used SMOTE–ENN as data balancing and Chi-square as the feature selection approach and the stacking ensemble approach to achieve the most effective results. The analysis proves that the proposed hybrid model is a novel solution for detecting cardiovascular disease compared to the existing models.

The proposed hybrid model attained the best classification accuracy of 97.8% and ROC-AUC score of 98.6%, demonstrating superior performance in the detection of cardiovascular disease. The proposed hybrid model successfully handles the major challenge of the imbalanced dataset using the hybrid data balancing approach named SMOTE–ENN. The hybrid SMOTE–ENN approach efficiently handles the class imbalance problem, resulting in improved model performance. The feature selection approach selects the highly ranked features to train the model, leading to less computational time and improved model performance in the detection of cardiovascular disease. A comparison of the proposed hybrid model with conventional methods like LR, DT, RFT, NB, and AdaBoost reveals that the proposed hybrid model shows superior performance compared to other traditional methods. In addition, the comparison highlights the efficiency and novelty of the proposed hybrid model over other existing models.

5 Conclusion

Recent advancements in AI and ML techniques have provided a new way of identifying patients suffering from various diseases and included in the current study to design a model for identifying patients suffering from cardiovascular disease. In this current research, a hybrid cardiovascular disease detection model is developed and evaluated using different performance evaluation metrics, including accuracy, sensitivity, specificity, F1 score, and ROC-AUC curve. The most crucial ML techniques are employed in the proposed model with the stacking ensemble approach, and their performance is computed using a tenfold CV method before and after the use of the hybrid sampling approach (SMOTE–ENN) and feature selection approach. The performance results show that the proposed hybrid model with a stacking approach with three base learners, RFT, K-NN, and AdaBoost and one meta-learner, LR, achieved the highest accuracy of 97.8%, 96.15% sensitivity, and 96.75% specificity and 98.6% of ROC-AUC score when compared with the other existing techniques and models. The proposed work not only detects cardiovascular disease but also provides the significant role of the SMOTE–ENN technique in enhancing the performance of the proposed hybrid model. From the results, it can be seen that the developed model reached an accuracy of 94.74% without the application of SMOTE–ENN and Chi-square as FS approach, which is enhanced to 97.8% after the application of SMOTE–ENN and Chi-square as FS approach. The proposed work can be extended in the future using more effective feature selection, sampling, and deep learning techniques, and the proposed hybrid model can also be implemented for other diseases.

6 Future Scopes

Though the proposed model demonstrates significant improvements in the detection of cardiovascular disease, it also suffers from some significant limitations. The proposed model works with a limited secondary dataset that is not significant in checking the robustness of the model. In addition, the model worked only with the clinical datasets; therefore, to make the model applicable in an IoT-based environment, it must be implemented on some sensor datasets. These limitations provide opportunities for further research in this direction; future work can focus on large datasets collected either clinically or from sensors to enhance their efficiency and to make them clinically actionable and acceptable. In addition, the efficiency of the proposed model can be further enhanced by incorporating more effective data balancing, feature selection, optimization approaches, and deep learning approaches.

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Data Availability The dataset is available with the corresponding author and available at individual request.

Declarations

Conflict of Interest The authors declare no competing interests.

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